Course notes

Circle and functions

On the trigonometric circle, the point identified by an angle x in radians has coordinates $(\cos x, \sin x)$. For every real x,

$$\cos^2 x + \sin^2 x = 1, \quad -1 \le \cos x, \sin x \le 1.$$

Cosine is even: $\cos(-x) = \cos x$. Sine is odd: $\sin(-x) = -\sin x$. The two functions are 2π-periodic.

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1

Derivatives and variations

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x.$$

$$(\sin(ax+b))' = a\cos(ax+b), \quad (\cos(ax+b))' = -a\sin(ax+b).$$

On $[-\pi,\pi]$, cosine increases on $[-\pi,0]$ of -1 to 1, then decreases on $[0,\pi]$ from 1 to -1. Sine decreases on $[-\pi,-\frac{\pi}{2}]$, increases on $[-\frac{\pi}{2},\frac{\pi}{2}]$, then decreases on $[\frac{\pi}{2},\pi]$.

Justification. Read the sign of sine and cosine on the circle and then apply the derivative link·variations. Periodicity extends the curves; it does not remove the need to study an interval of length sufficient.

Equations and inequalities

If $|a|>1$, $\cos x = a$ and $\sin x = a$ have no real solution. For a known angle θ,

$$\cos x = \cos\theta \iff x = \theta + 2k\pi \text{ or } x = -\theta + 2k\pi,$$

$$\sin x = \sin\theta \iff x = \theta + 2k\pi \text{ or } x = \pi - \theta + 2k\pi, \quad k \in \mathbb{Z}.$$

The solutions are then restricted to the required interval and duplicates are removed.

Examples on $[-\pi,\pi]$. $\cos x = \frac{1}{2}$ gives $x = \pm\frac{\pi}{3}$; $\sin x = \frac{1}{2}$ gives $x = \frac{\pi}{6}$ or $\frac{5\pi}{6}$. $\cos x \le \frac{1}{2}$ gives $[-\pi,-\frac{\pi}{3}]\cup[\frac{\pi}{3},\pi]$; $\sin x \ge \frac{1}{2}$ gives $[\frac{\pi}{6},\frac{5\pi}{6}]$.

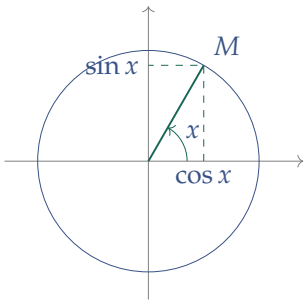
A trigonometric composite

To solve $\cos(2x) = \frac{1}{2}$ on $[-\pi,\pi]$, set $t = 2x \in [-2\pi, 2\pi]$, solve for t, then divide by 2. We get $x \in \{-\frac{5\pi}{6}, -\frac{\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}\}$.

Optimisation. For $f(x) = \sin x + \frac{1}{2}\sin(2x)$ on $[0,\pi]$, $f' = \cos x + \cos(2x) = 2\cos^2 x + \cos x - 1 = (2\cos x - 1)(\cos x + 1)$. The function increases up to $\frac{\pi}{3}$ then decreases; maximum $\frac{3\sqrt{3}}{4}$.

Duplication identities used in exercises. For every real number x, $\sin(2x) = 2\sin x\cos x$ and $\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1$. These identities are recalled here so that their use in the problems is not left implicit.

Curve symmetries. The parity of cosine corresponds to a symmetry with respect to the axis of y-axis; the oddness of sine to a point symmetry about the origin. Periodicity extends each curve by translation by the vector $(2\pi, 0)$.



The coordinates of the point on the circle are $(\cos x, \sin x)$.

From the circle to the study of a function

5.1 One directed angle, two coordinates

First use: S75.

Definition · Sine and cosine

On the unit circle oriented anticlockwise, the point at angle x has coordinates $(\cos x; \sin x)$. Add $2 \quad k\pi$ with $k \in \mathbb{Z}$ gives the same point. A negative measure corresponds in the opposite direction; it is not a negative length.

Worked example · Reading an equation and an inequality

On $[-\pi; \pi]$, $\cos x = 1/2$ corresponds to the two points with x-coordinate 1/2: $x = \pm\pi/3$. The inequality $\cos x \leq 1/2$ selects the arc to the left of the vertical line x = 1/2: $[-\pi; -\pi/3] \cup [\pi/3; \pi]$. The endpoints belong to the solution because the inequality is non-strict.

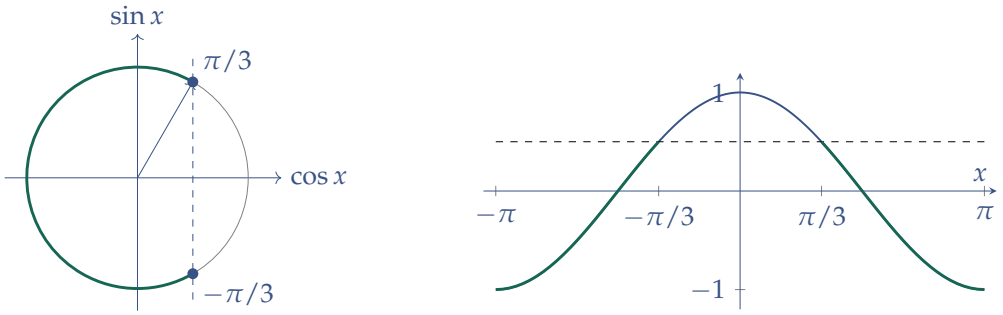


Figure 5.1. Same condition $\cos x \leq 1/2$: arc to the left of the vertical line on the circle, curve below level 1 /2 à line.

A1 · Immediate application Solve $\sin x = 1/2$ on $[0; 2\pi[$, then $\sin x \geq 1/2$ on that same interval. Show the solutions on a circle and on a curve.

5.2 Reading amplitude and period

First use: S76.

Worked example · A shifted oscillation

For $h(t) = 3 + 2\sin(4t)$, the values lie between 1 and 5: the mean level is 3, the amplitude is 2. The smallest positive period is 2 $\pi/4 = \pi/2$. The number 2 π remains a period, but is not the smallest.

A2 · Partial guidance For $g(t) = 1 - 3\cos(2t)$, give the minimum, maximum, range and largest smallest positive period. Find the values of t in $[0; \pi]$ where the extrema are attained.

A3 · Refute "The smallest positive period of $\sin^2 x$ is 2 π, such as that of $\sin x$." Show than a translation of π leaves the function unchanged; explain why $\pi/2$ is not a period.

5.3 The sign comes from the circle

First use: S78.

Property · Derivatives and antiderivatives

The derivatives are $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$. For a composite, multiply by the inner derivative. An antiderivative of $\cos(ax+b)$ for $a \neq 0$ is $\sin(ax+b)/a$; an antiderivative of $\sin(ax+b)$ is $-\cos(ax+b)/a$.

Worked example · Variation on an interval

For $f(x) = x + \cos x$ on $[0;\pi]$, $f' = 1 - \sin x \geq 0$, with equality only at $\pi/2$. The function is strictly increasing: it strictly increases on each side of this isolated point and remains continuous. Moreover $f'' = -\cos x$ changes sign at $\pi/2$, so the curve has an inflection at $(\pi/2;\pi/2)$. A horizontal tangent is not necessarily an extremum.

A4 · Choice of strategy Study $f(x) = x - \sin x$ on $[0;\pi]$: derivative, variations, convexity. Deduce $\sin x \leq x$ on this interval. The formula $f'(0) = 0$ imply a change of variation?

5.4 An identity serving an area

First use: S79.

Worked example · Area of a rectangle

A point in the first quadrant of the unit circle has coordinates $(\cos\theta;\sin\theta)$, $0 \leq \theta \leq \pi/2$. The rectangle symmetric about the axes has area $A(\theta) = 4\sin\theta\cos\theta$. Since $(\sin\theta - \cos\theta)^2 \geq 0$ and $\sin^2\theta + \cos^2\theta = 1$, we have $2\sin\theta\cos\theta \leq 1$, therefore $A \leq 2$. Equality requires $\theta = \pi/4$: the optimal rectangle is a square.

A5 · Synthesis Determine by differentiation the maximum of $4\sin\theta\cos\theta$ on $[0;\pi/2]$. Recover the previous conclusion, then calculate $\int_0^{\pi/2} \sin\theta\cos\theta\,\mathrm{d}\theta$.

A6 · Independent problem Prove that $\cos x = x$ has a unique solution on $[0;\pi/2]$. Bound it between 0.7 and 0.8 using the calculator in radian mode. Suggest a method to improve the precision.

A first step: Study the difference of the two sides and separate existence, uniqueness and approximation.

One minute, spoken aloud. Explain how the same set of solutions appears as arcs on the circle and intervals on the x-axis of the curve.

Delayed review · S84 then S90. Mix limits, differentiation, equations, areas and trigonometry in same study; redo a solution on another interval, checking the endpoints.

Investigation

Height of a point on a wheel - 35 min.

A wheel turns at constant speed in a vertical plane. Its radius is 2 m and its centre is is 3 m above the ground. A point on the wheel is at its lowest position at time $t = 0$. The wheel completes one revolution in 8 s. Its height is modelled by $h(t) = 3 - 2\cos\left(\frac{\pi t}{4}\right)$, where t is expressed in seconds and $h(t)$ in metres.

1. Check that the model gives the stated initial height, a period of 8 s and the heights expected minimum and maximum.
2. Calculate $h'(t)$, then study the variation of h on $[0;8]$.
3. Determine all the times $t \in [0;8]$ for which the height is 4 m.
4. Determine the total time, during this turn, for which the point is at least 4 m above the ground.